

03 – Case MOBILITY SHIFT

Stage 1 – Baseline Network Diagnostics & Growth-Aware Forecasting

 28 February 2026 | 11:00 AM

1. Executive Context

Dubai RTA manages a multi-corridor bus network serving:

- CBD corridors (Downtown, Business Bay, Deira)
- Coastal tourism belt (Marina)
- Residential zones (Al Qusais, International City)
- Industrial corridor (Jebel Ali)

Between 2022 and mid-2025:

- Urban population expanded steadily
- Tourism exhibited strong seasonal variation
- Road congestion intensified gradually
- Corridor-level demand divergence increased

The Board mandate:

“Align demand, corridor efficiency, and fleet allocation under accelerating urban growth.”

This is no longer a capacity estimation problem.

It is a system efficiency and allocation optimization problem under growth and congestion pressure.

2. Operating Environment Characteristics

A. Demand Environment

- Winter uplift (Nov–Mar)
- Summer moderation (Jun–Aug)
- Event-driven mobility shifts
- Weekday/weekend divergence
- Non-uniform corridor growth

B. Network Structure

Routes categorized as:

- City
- Express
- Feeder
- Intercity

Stop-level demand varies by:

- Urban Zone
- Route Type
- Day Type
- Congestion Level

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C. Congestion Interaction

Congestion affects:

- Average speed
- Effective capacity
- Load redistribution
- Reliability

Full stationarity must not be assumed.

3. Data Provided

A. Network Structure

- Bus_Routes.csv
- Bus_Stops.csv
- Route_Stop_Mapping.csv

B. Ridership Data (Training)

- Train_Ridership_2022_to_2025H1.csv

Derived metric (recommended): [Total_Pax = Boarding_Count + Alighting_Count]

C. Traffic Data

- Train_Traffic_2022_to_2025H1.csv

Datasets must be merged using:

- Route_ID
- Stop_ID
- Date

Incorrect relational merging may materially distort corridor analysis.



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4. Advisory Mandate

You must:

1. Diagnose multi-year demand dynamics (2022–H1 2025).
2. Detect seasonality and weekday/weekend divergence.
3. Identify corridor-level structural imbalance.
4. Model congestion–demand interaction.
5. Forecast route-level daily demand for: **July 1 – December 31, 2025.**
 - a) Identify:
 - Overload risk corridors
 - Underutilized capacity
 - Emerging imbalance
 - b) Propose operational adjustments:
 - Fleet reallocation
 - Headway modification
 - Corridor prioritization

5. Stage 1 Expectations

By 7:00 PM, you should have:

- Growth decomposition
- Seasonal insight
- Route-type comparison
- Congestion elasticity understanding
- Baseline forecast
- Initial allocation strategy

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Stage 2 – Structural Break: Metro Phase 2 Introduction

 28 February 2026 | 7:00 PM

1. Structural Regime Shift

Metro Phase 2 commenced July 1, 2025.

Parallel impact corridors include:

- CBD_Downtown
- CBD_BusinessBay
- Core_Deira
- Select residential corridors

System effects observed:

- Feeder contraction
- Express redistribution
- CBD overlap reduction
- Congestion redistribution

This represents a structural break.

2. New Data Released

- Shock_Ridership_2025_Q3.csv
- Shock_Traffic_2025_Q3.csv

Covering July 1 – September 30, 2025.

This period was not included in Stage 1 modeling.

3. Stage 2 Mandate

You must:

A. Detect Structural Break

Quantify:

- Stage 1 forecast vs Q3 actual deviation
- Route-level divergence magnitude
- Shift classification:
 - Level shift
 - Volatility shift
 - Elasticity shift
 - Congestion-mediated shift

03 – Case MOBILITY SHIFT**B. Recalibrate Forecast**

Using:

- 2022–H1 2025
- Q3 2025

Produce revised forecasts for:

October 1 – December 31, 2025

Explicitly state:

- Temporary shock vs regime change
- Elasticity adjustment logic
- Uncertainty revision

C. Operational Risk Reassessment

Identify:

- Overload corridors
- Underutilized routes
- Feeder viability
- Express sustainability

D. Fleet Reallocation Strategy

Propose:

- Headway revisions
- Fleet redeployment
- Corridor rationalization
- Volatility buffer approach

Quantify:

- Passenger redistribution
- Efficiency impact
- Congestion interaction

03 – Case MOBILITY SHIFT**Stage 3 – Strategic Accountability & Stabilized Regime Evaluation** 01 March 2026 | 1:00 AM**1. Out-of-Time Operational Data Release**

Provided:

- OutOfTime_Ridership_2025_Q4.csv
- OutOfTime_Traffic_2025_Q4.csv

Covering October 1 – December 31, 2025.

This reflects:

- Partial fleet rationalization
- Express frequency moderation
- Feeder adjustments
- Persistent winter uplift
- Stabilized redistribution

The network neither reverted nor retained full shock intensity.

It entered a redistributed equilibrium.

2. Stage 3 Mandate – Accountability Audit

This stage evaluates structural judgment across all stages.

It is not merely forecast validation.

3. Required Analysis**A. Forecast Performance Audit**

Compare:

- Stage 1 forecast vs Q3 actual
- Stage 2 forecast vs Q4 actual

Report:

- Route-level MAPE
- Route-type aggregated error
- Directional bias
- Variance misestimation

Identify:

- Overreaction vs underreaction
- Structural permanence misclassification

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B. Strategic Alignment Evaluation

Assess:

- Did Stage 2 fleet reallocation align with Q4 stabilization?
- Were adjustments excessive or insufficient?
- Did elasticity assumptions persist or decay?

C. Elasticity & Substitution Diagnosis

Determine:

- Metro substitution elasticity magnitude
- Congestion elasticity persistence
- Route-type structural rebalancing

D. 2026 Forward Strategy

Propose a stabilized 2026 design:

- Feeder redesign
- Express capacity calibration
- CBD rationalization framework
- Volatility buffer mechanism

Quantify:

- Passenger redistribution
- Efficiency gain
- Risk exposure under future infrastructure expansion

4. Final Submission (9:00 AM)

All Stage 1, 2, and 3 analyses must be integrated into a single final submission.

No partial submissions will be evaluated.